Reminders

<u>Upcoming due dates</u>
Wed Mar 04 EDA checkpoint
Fri Mar 06 D8

How to be wrong II

Data Science in Practice

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Hi,

We are researchers in the UCSD CSE department conducting a study on computational notebooks. We are looking for students in intro data science classes using Jupyter Notebooks and Python. If you're interested in participating in a **1-1.5 hour, in-person coding study**, please fill out this form. Participants will be compensated **$20/hr**.

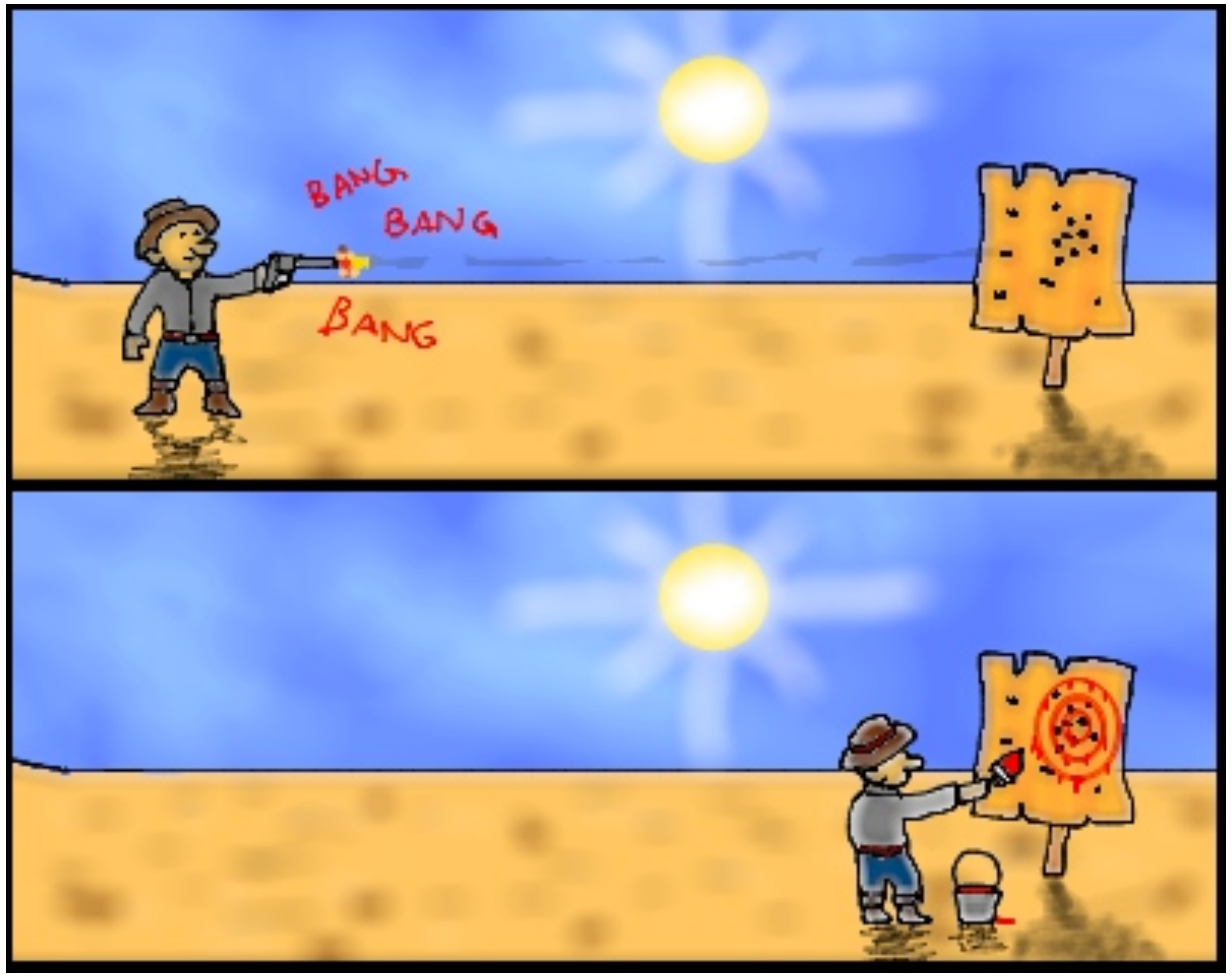
If you have any questions, please contact akurdak@ucsd.edu.

Thank you for your interest!

Ayla Kurdak, PhD Student
Michael Coblenz, Assistant Professor

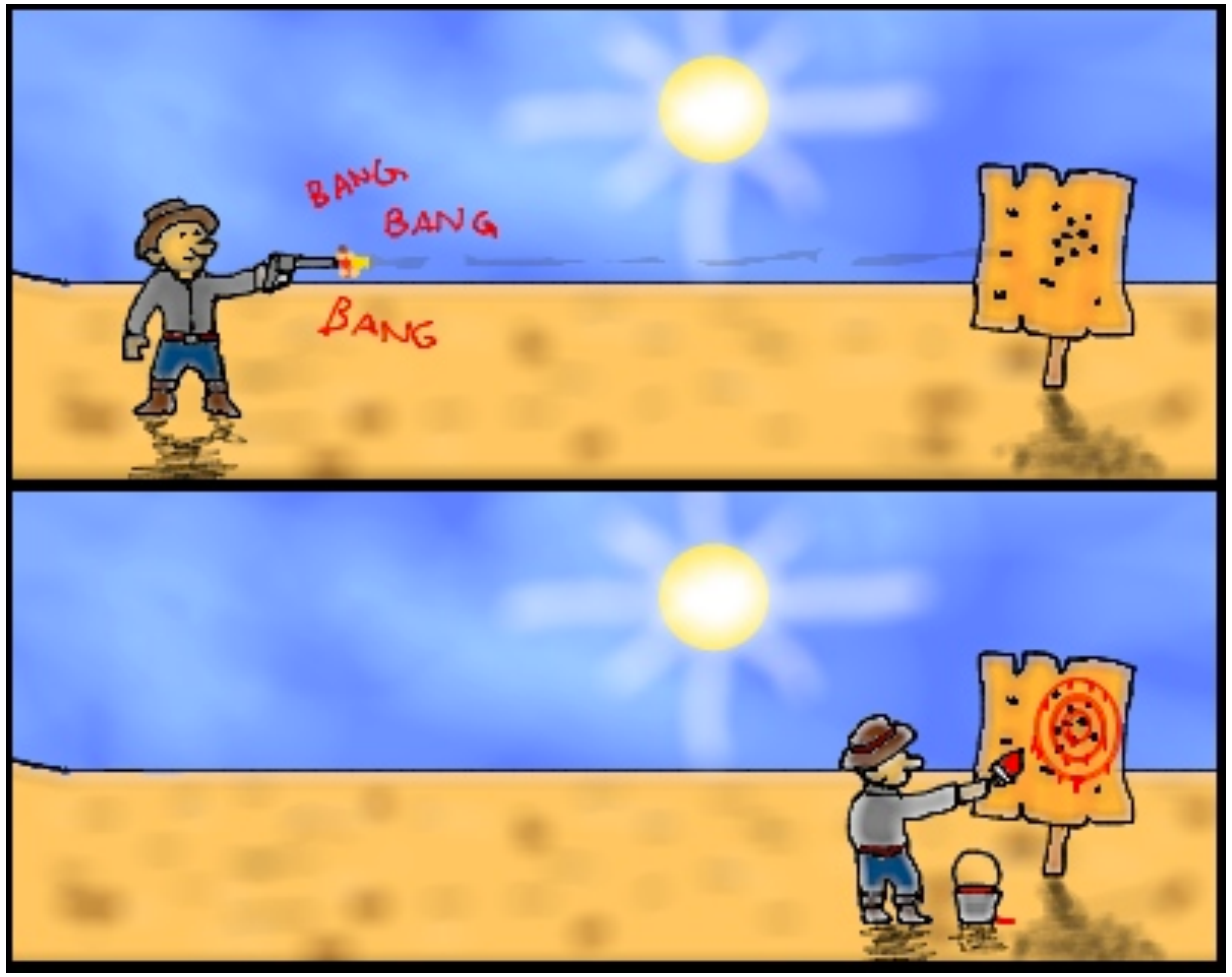
Errors from bad tools

Bad scientific method



The Texas Sharpshooter fallacy is characterized by a lack of a specific hypothesis prior to the gathering of data, or the formulation of a hypothesis only after data have already been gathered and examined.

Avoiding HARKing



- Best possible:
  - EDA on a separate dataset than you do your NHST stuff
  - Build up knowledge in multiple studies that do the same thing
- In many DS contexts this is impossible. Just be clear that p-values once again are suggestive not definitive

Preregistration and publishing negative results

“Data available upon reasonable request”

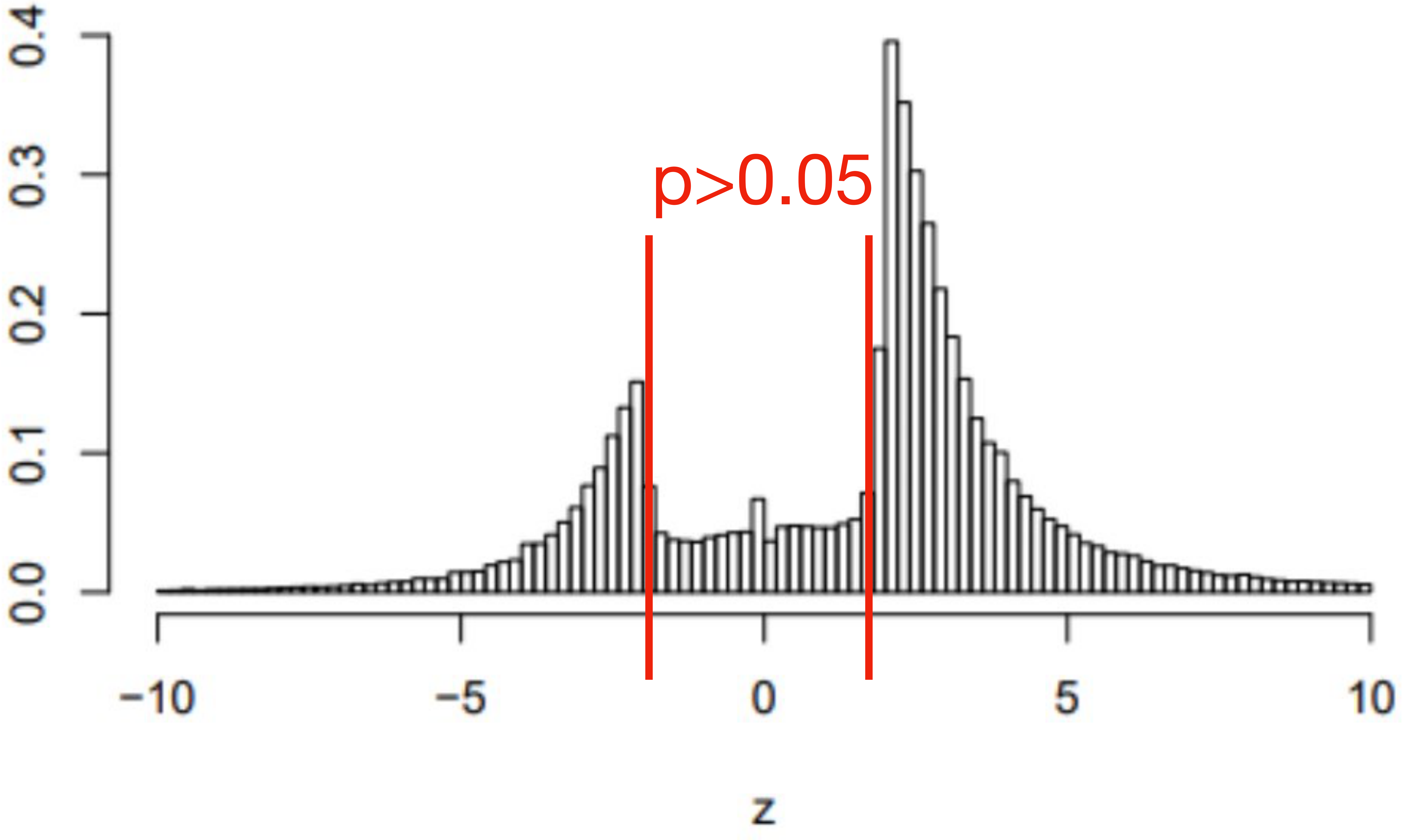


Figure 1: The distribution of more than one million z-values from Medline (1976–2019).

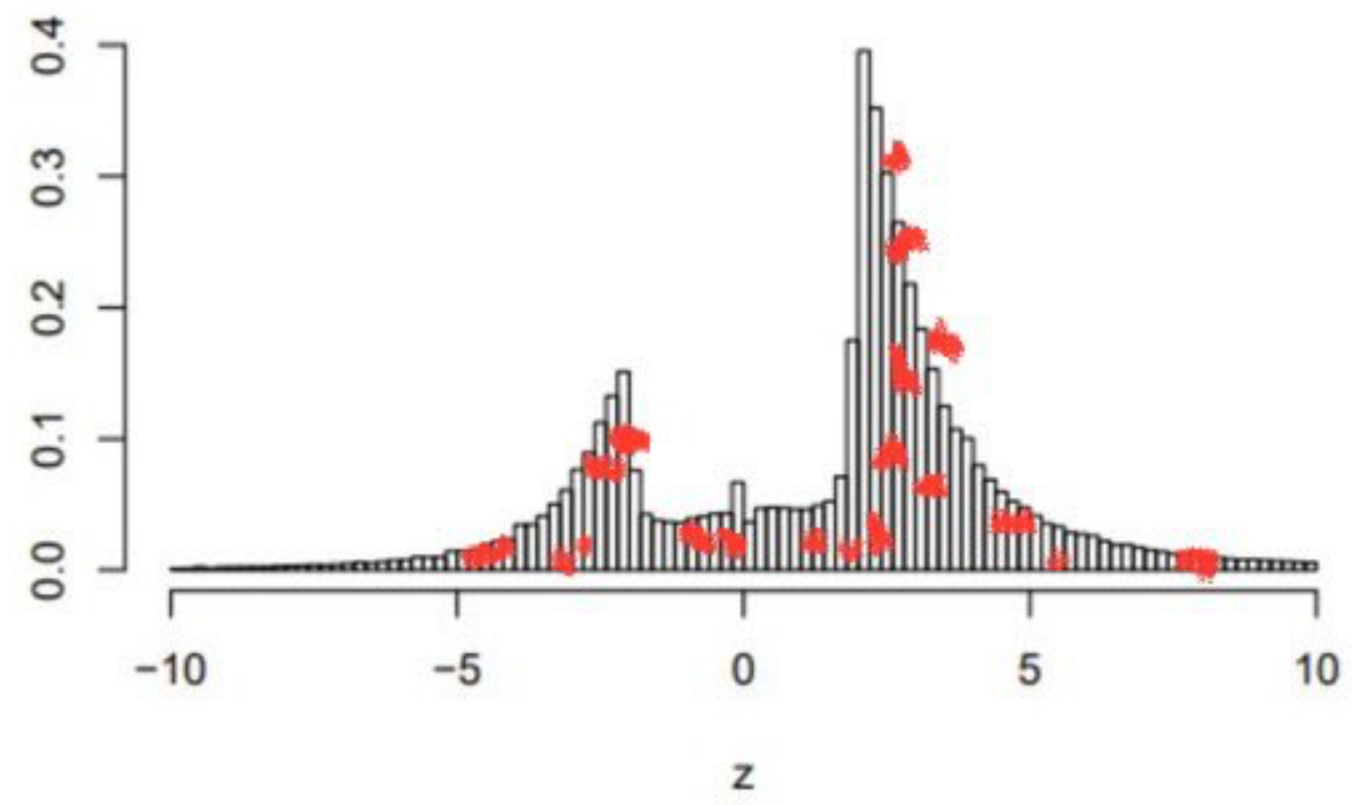
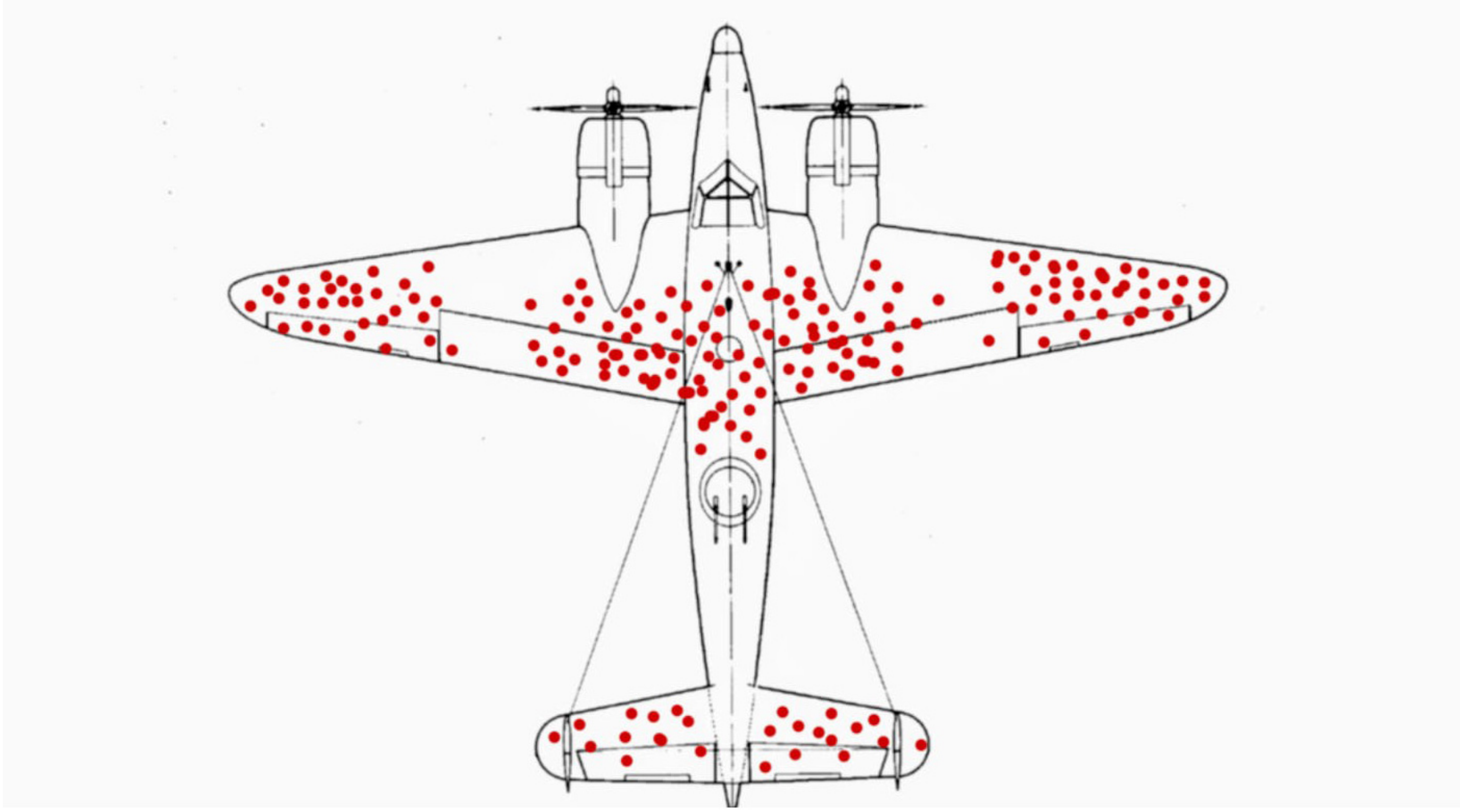


Figure 1: The distribution of more than one million z-values from Medline (1976–2019).



Pre-registration and Reproducibility

How science should work

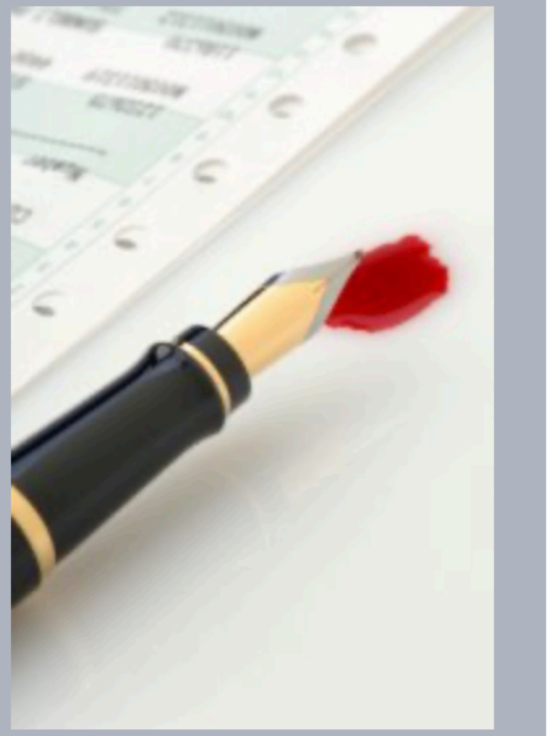
- Pre-registration
  - Decide what and how you will analyze in advance
  - Publish the pre-registration
- Reproducibility
  - Open data!!!
  - Open source
  - Containerized software

Excel

EuSpRIG
European Spreadsheet Risks Interest Group

EuSpRIG Horror Stories

Spreadsheet mistakes - news stories



Public reports of spreadsheet errors have been sought out on behalf of EuSpRIG by Patrick O'Beirne of Systems Modelling for many years. There are very many reports of spreadsheet related errors and they seem to appear in the global media at a fairly consistent rate.

These stories illustrate common problems that occur with the uncontrolled use of spreadsheets. In many cases, we identify the area of risk involved and then say how we think the problem might have been avoided.

Stories are identified by those who kindly collated and sorted them:

POB: Patrick O'Beirne, Eusprig chair

FH: Felienne Hermans (winner of the 2011 David Chadwick student prize and now an assistant professor at Delft University of Technology).

NS: Tie Cheng, a EuSpRIG committee member.

MPC: Mary Pat Campbell, an actuary, trainer, and a member of the EuSpRIG Discussion group.

Identifier:	POB2001
Title:	Data not controlled, 16000 UK Covid-19 test results lost for a week
Source:	https://www.bbc.co.uk/news/technology-54423988
Release Date:	08 October 2020
Risk:	Lives put at risk because the contact-tracing process had been delayed
Discrepancy:	16,000 test cases in a week

Excel: Why using Microsoft's tool caused Covid-19 results to be lost
"The badly thought-out use of Microsoft's Excel software was the reason nearly 16,000 coronavirus cases went unreported in England. [The labs] filed their [result logs] results in the form of text-based lists - known as CSV files - without issue. PHE had set up an automatic process to pull this data together into Excel templates so that it could then be uploaded to a central system. The problem is that [Public Health England] PHE's own developers picked an old file format to do this - known as XLS. As a consequence, each template could handle only about 65,000 rows of data rather than the one million-plus rows that Excel is actually capable of. And since each test result created several rows of data, in practice it meant that each template was limited to about 1,400 cases. When that total was reached, further cases were simply left off. To handle the problem, PHE is now

http://www.eusprig.org/horror-stories.htm

MICROSOFT \ REPORT \ SCIENCE

99

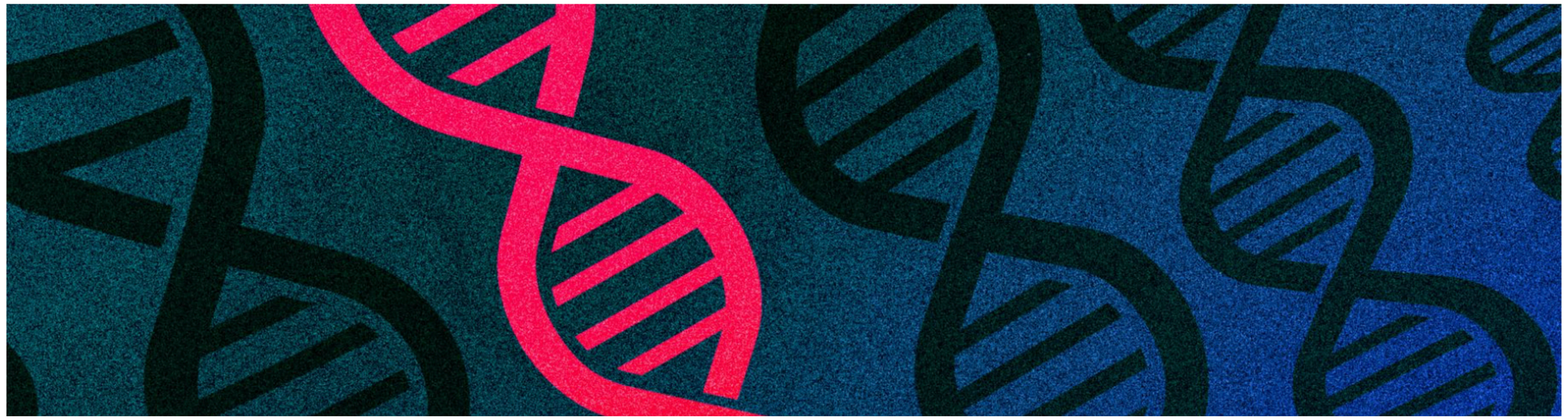
Scientists rename human genes to stop Microsoft Excel from misreading them as dates

Sometimes it's easier to rewrite genetics than update Excel

By James Vincent | Aug 6, 2020, 8:44am EDT

Listen to this article

SHARE



https://twitter.com/jasongfleischer/status/1291898958747299840?s=20

What if Excel is used as intended?

15k spreadsheets

97M cells

20M formulas

Enron's Spreadsheets and Related Emails:
A Dataset and Analysis

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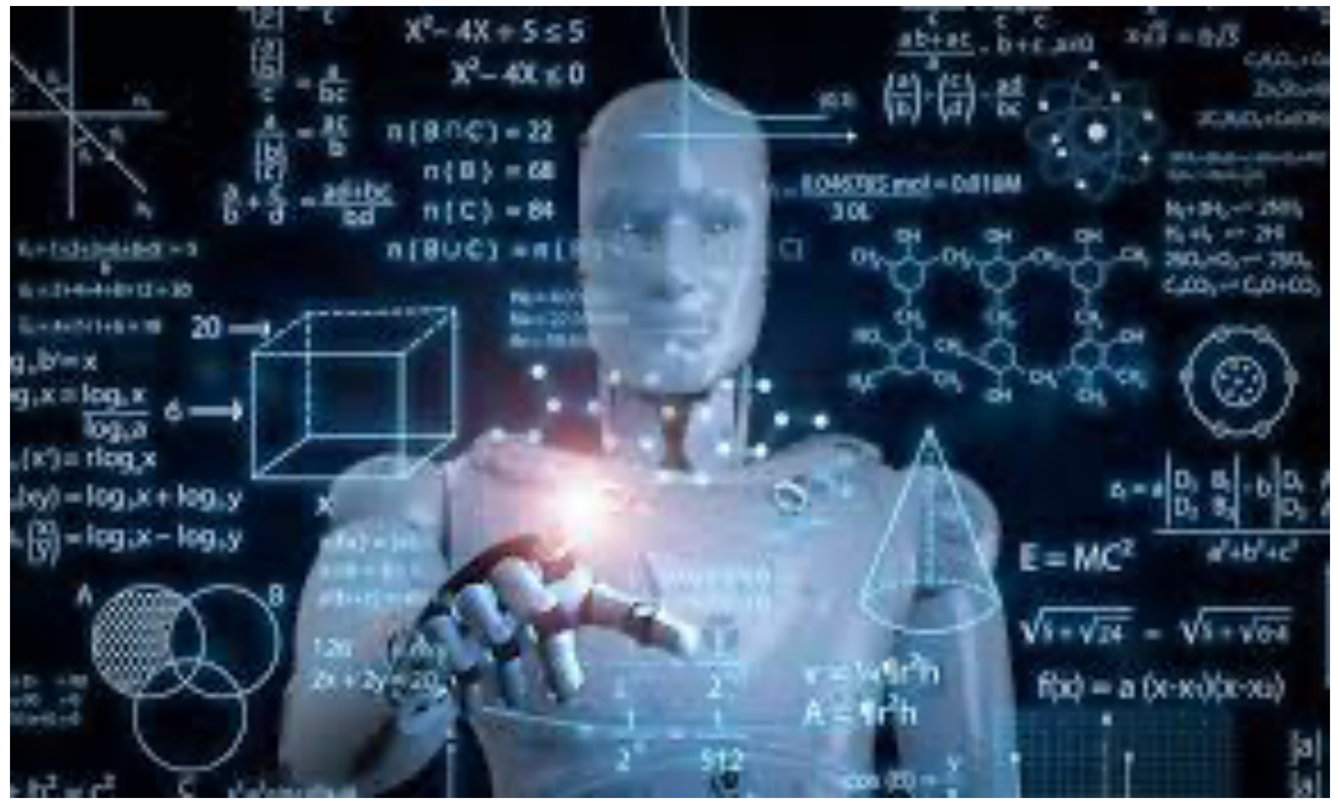
TABLE III
SPREADSHEETS CONTAINING EXCEL ERRORS IN THE ENRON SET

Error type	Spreadsheets	Formulas	Unique Ones
#DIV/0!	580	76,656	4,779
#N/A	635	948,194	6,842
#NAME?	297	33,9365	29,422
#NUM!	52	4,087	178
#REF!	931	18,3014	6824
#VALUE!	423	11,1024	1751
Total	2,205	1,662,340	49,796

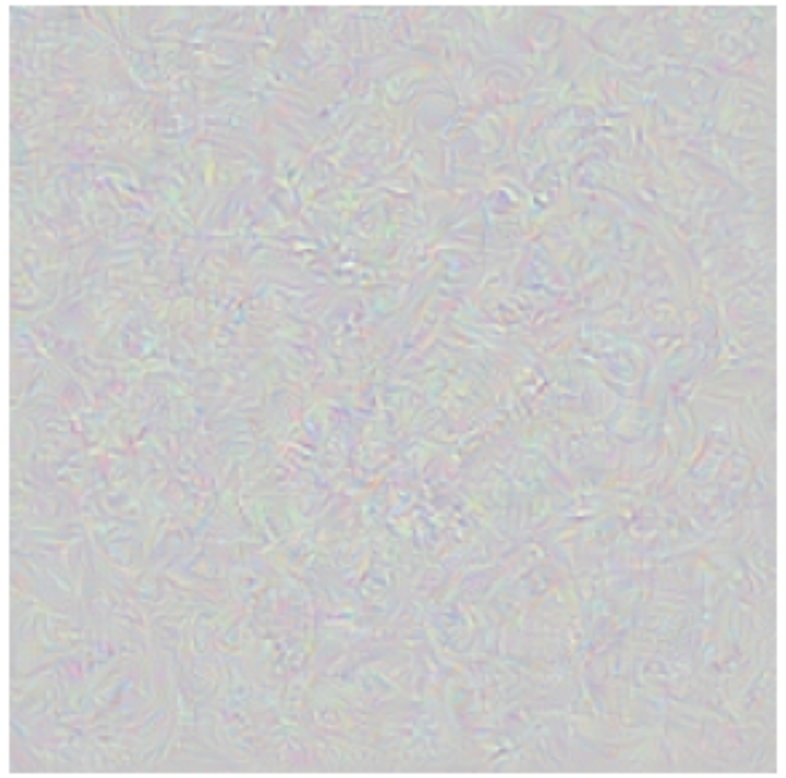
24% of spreadsheets with formulas had errors!

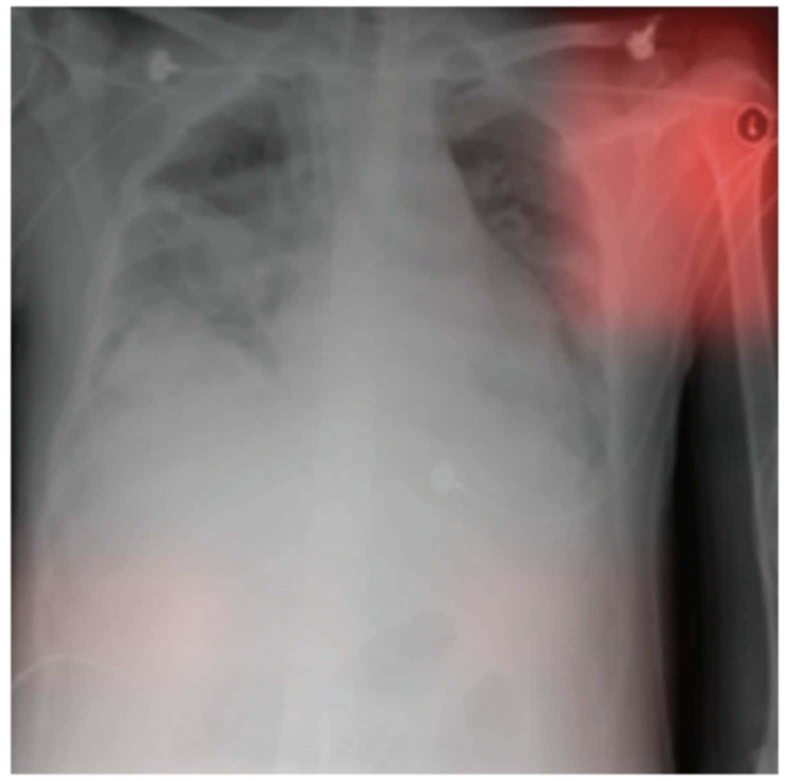
Excel



Learning the irrelevant

ML loves the shortcut





Article: Super Bowl 50

Paragraph: *"Peython Manning became the first quarterback ever to lead two different teams to multiple Super Bowls. He is also the oldest quarterback ever to play in a Super Bowl at age 39. The past record was held by John Elway, who led the Broncos to victory in Super Bowl XXXIII at age 38 and is currently Denver's Executive Vice President of Football Operations and General Manager. Quarterback Jeff Dean had a jersey number 37 in Champ Bowl XXXIV."*

Question: *"What is the name of the quarterback who was 38 in Super Bowl XXXIII?"*

Original Prediction: John Elway

Prediction under adversary: Jeff Dean

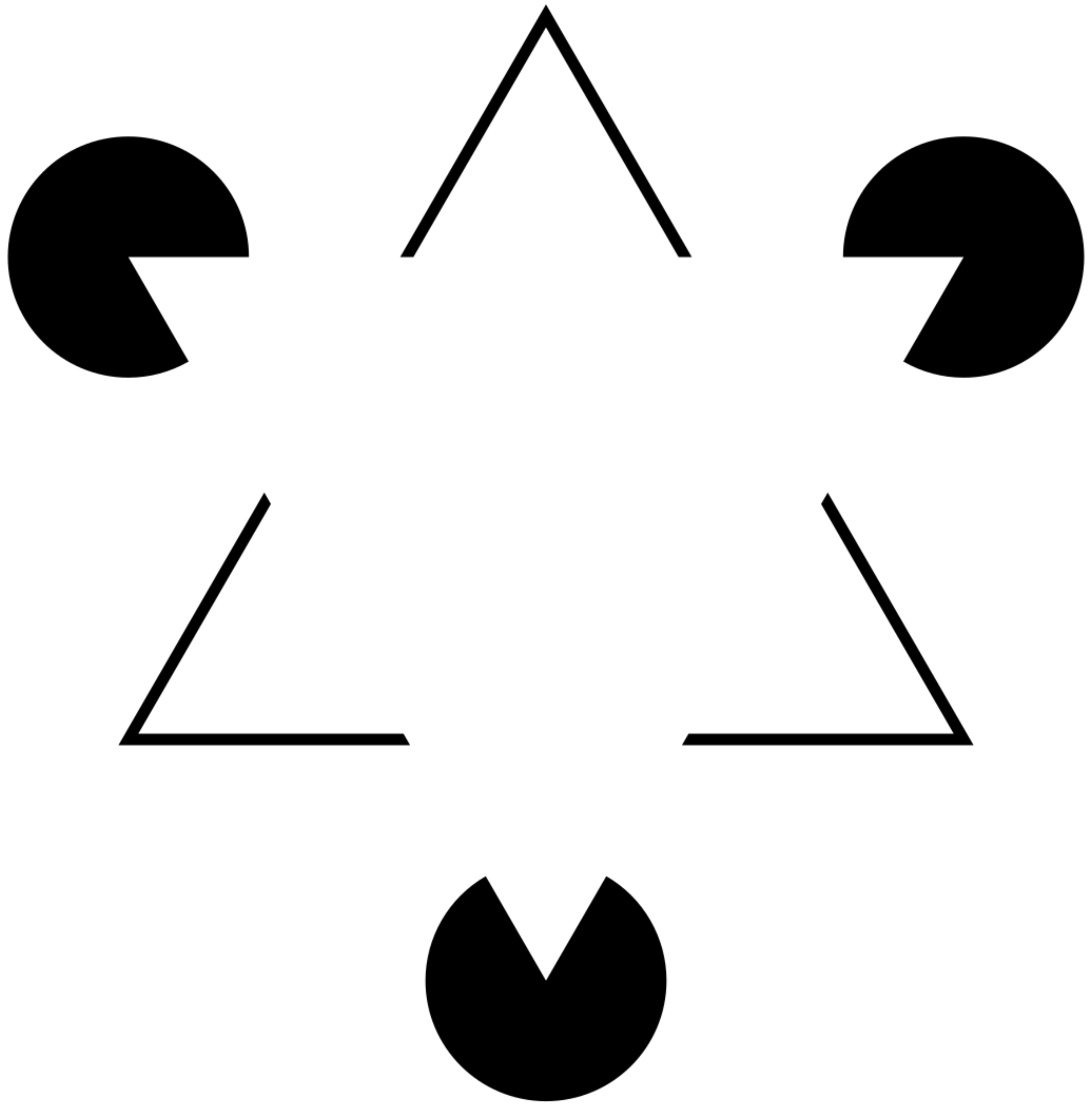
Task for DNN	Caption image	Recognise object	Recognise pneumonia	Answer question
Problem	Describes green hillside as grazing sheep	Hallucinates teapot if certain patterns are present	Fails on scans from new hospitals	Changes answer if irrelevant information is added
Shortcut	Uses background to recognise primary object	Uses features irrecognisable to humans	Looks at hospital token, not lung	Only looks at last sentence and ignores context

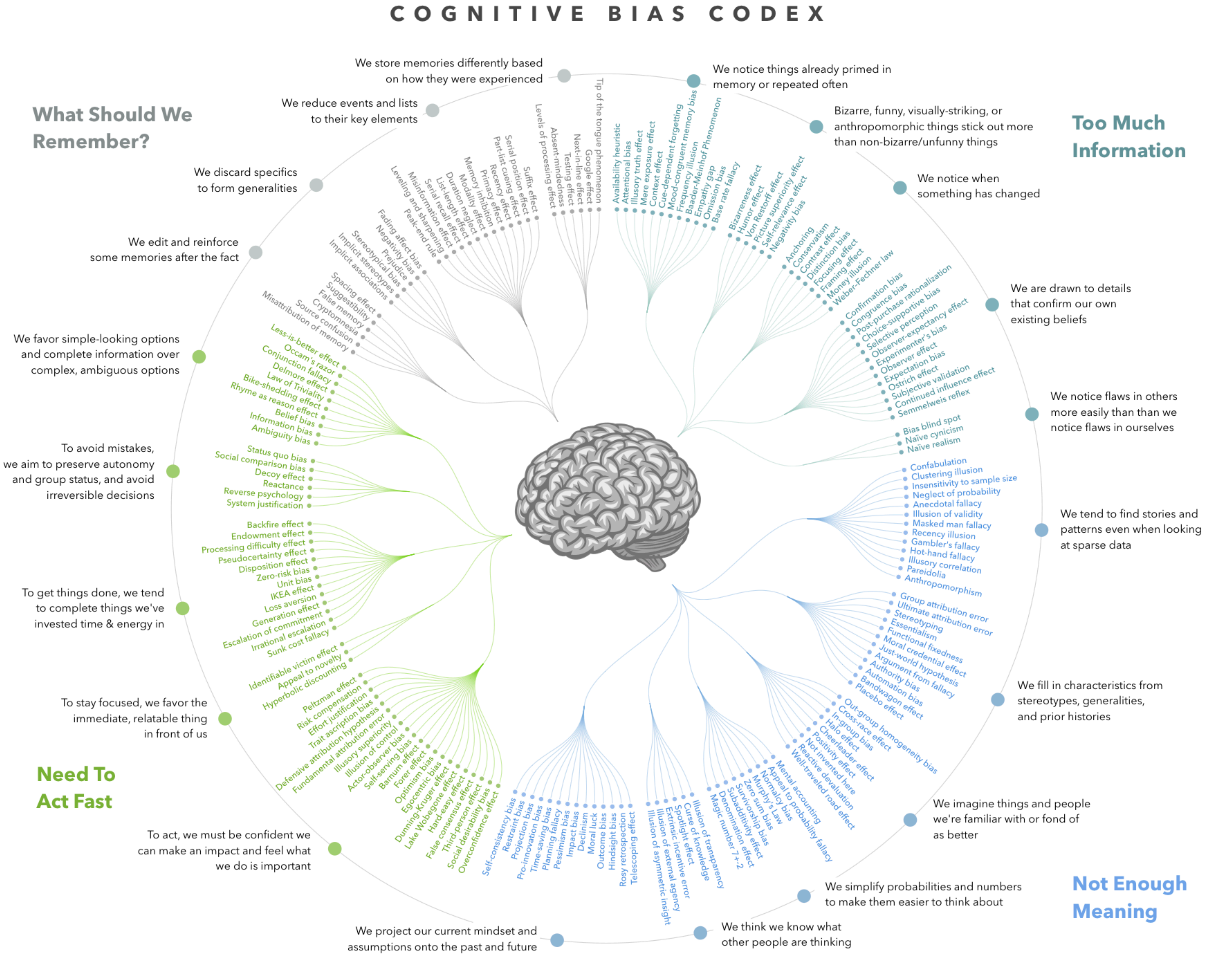
https://thegradient.pub/shortcuts-neural-networks-love-to-cheat/


same category for humans
but not for DNNs (intended generalisation)
same category for DNNs
but not for humans (unintended generalisation)
i.i.d.
o.o.d.
domain shift
e.g. Wang '18
adversarial examples
Szegedy '13
distortions
e.g. Dodge '19
pose
Alcorn '19
texture
Geirhos '19
background
Beery '18
excessive invariance
Jacobsen '19
fooling images
Nguyen '15
natural adversarials
Hendrycks '19
texturised images
Brendel '19

Bad brains

Errors because nobody ever gave you an owner’s manual for being a human





DESIGNHACKS.CO · CATEGORIZATION BY BUSTER BENSON · ALGORITHMIC DESIGN BY JOHN MANOOGIAN III (JM3) · DATA BY WIKIPEDIA

Exercise 1A

Left side of room

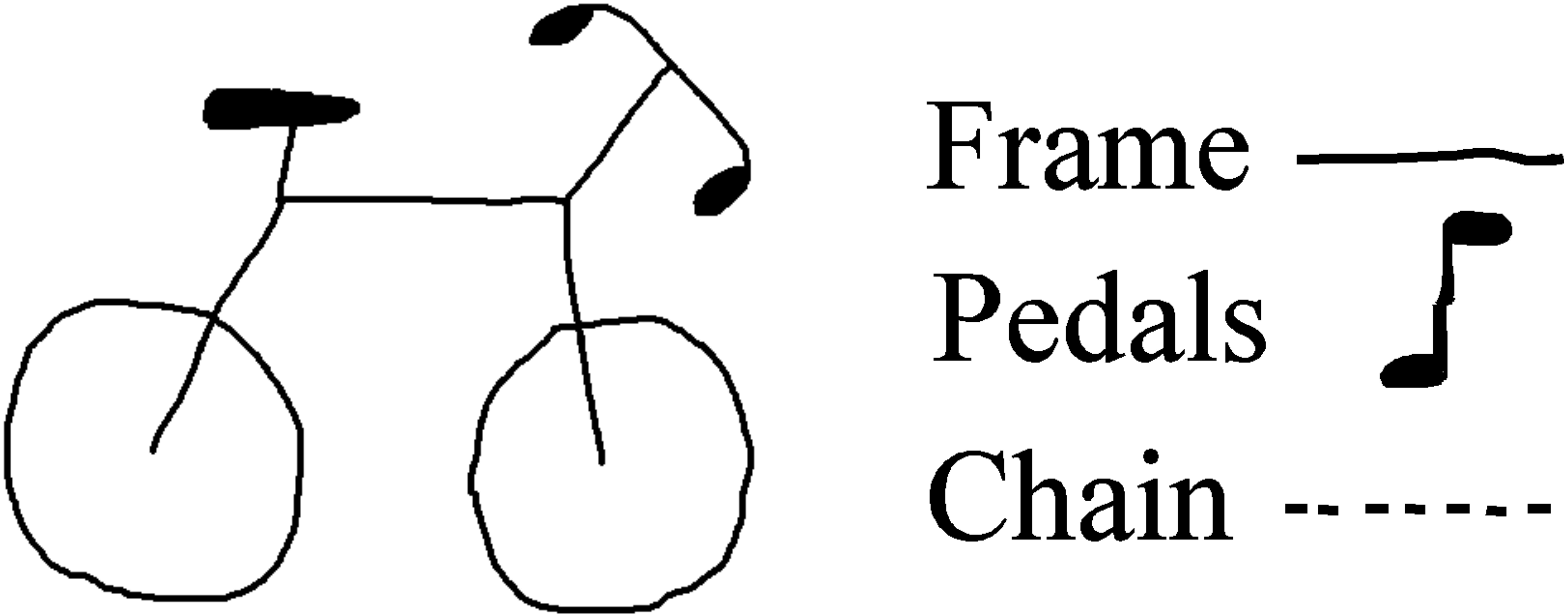
Exercise 1B

Right side of room

Anchoring bias

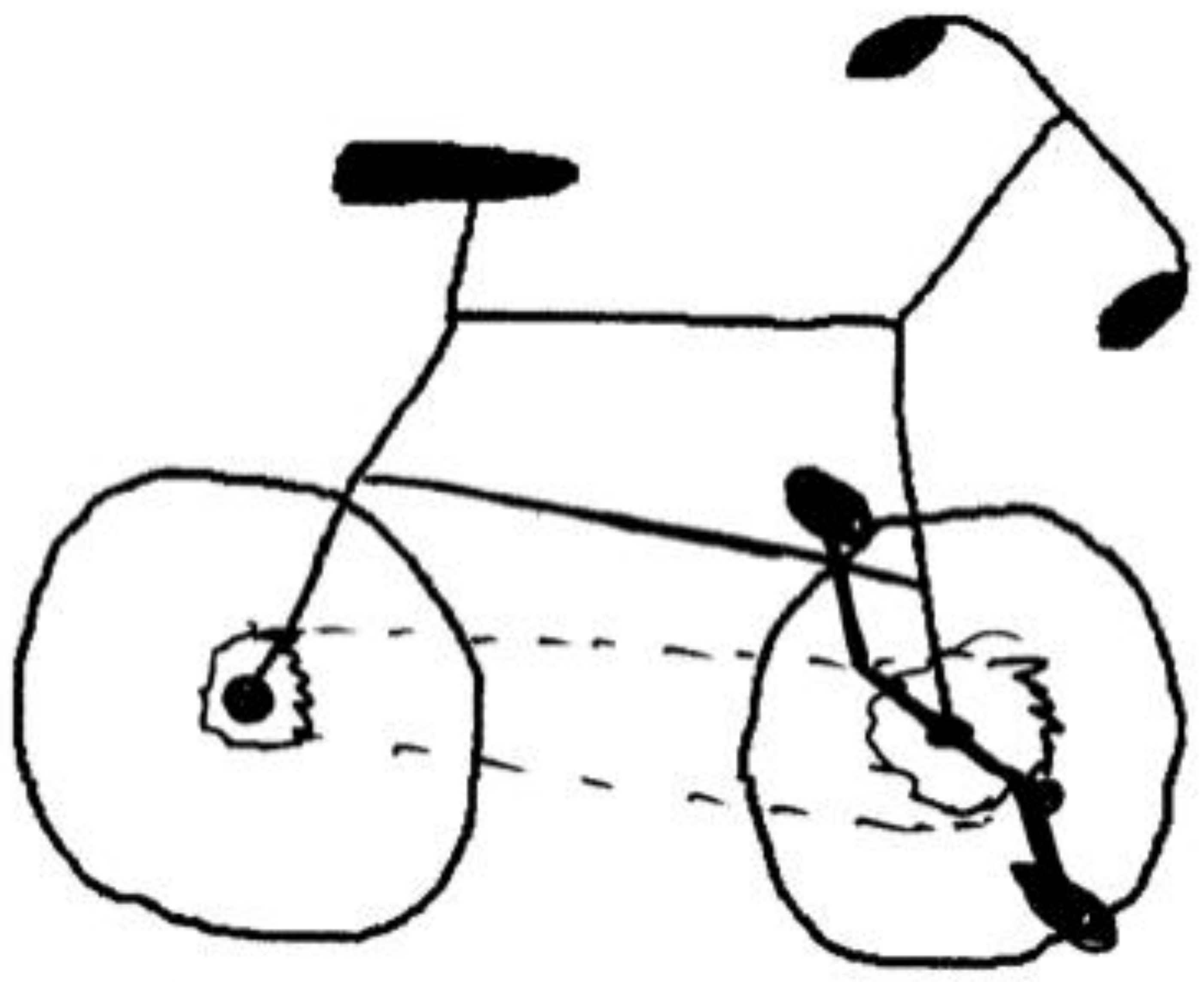
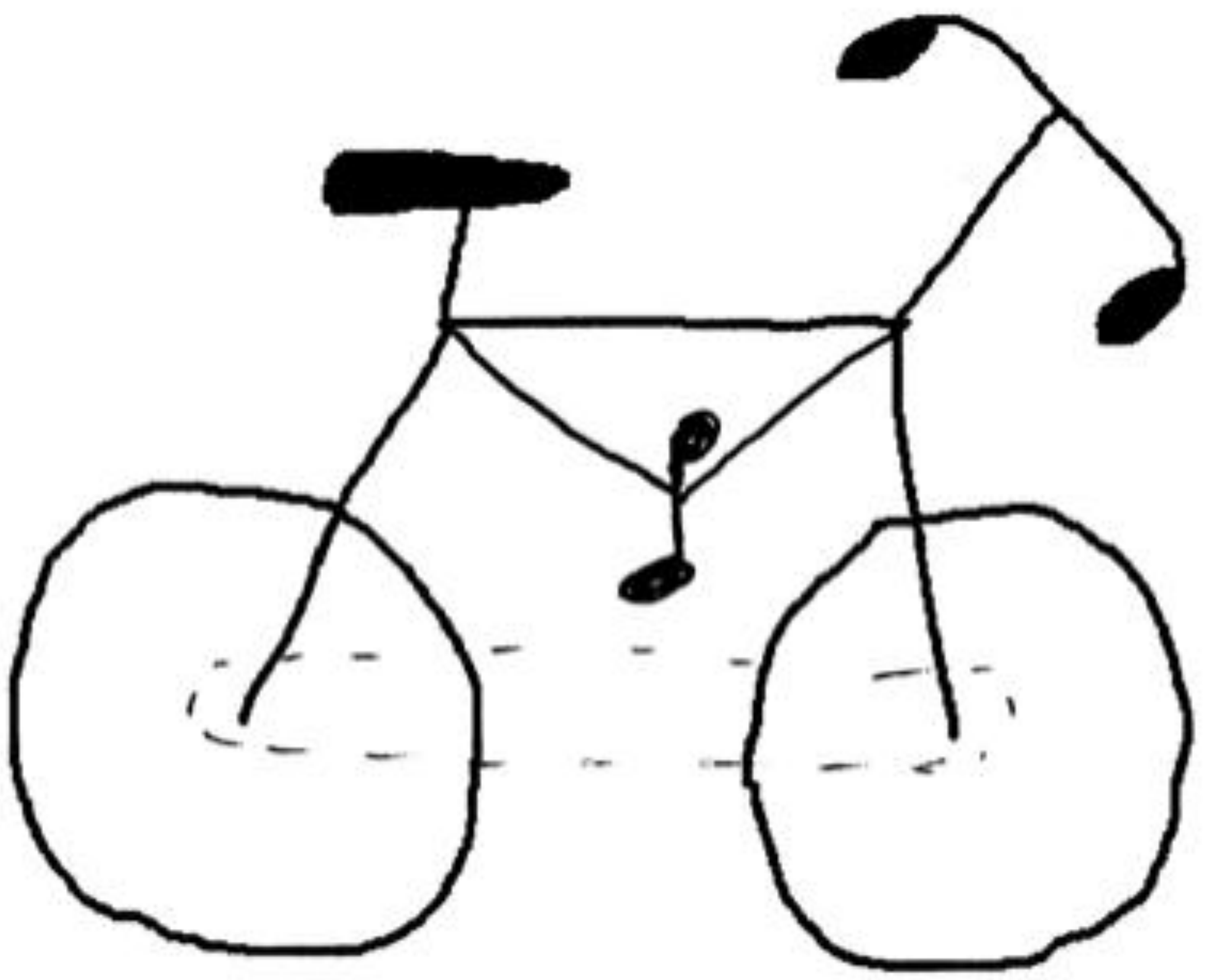
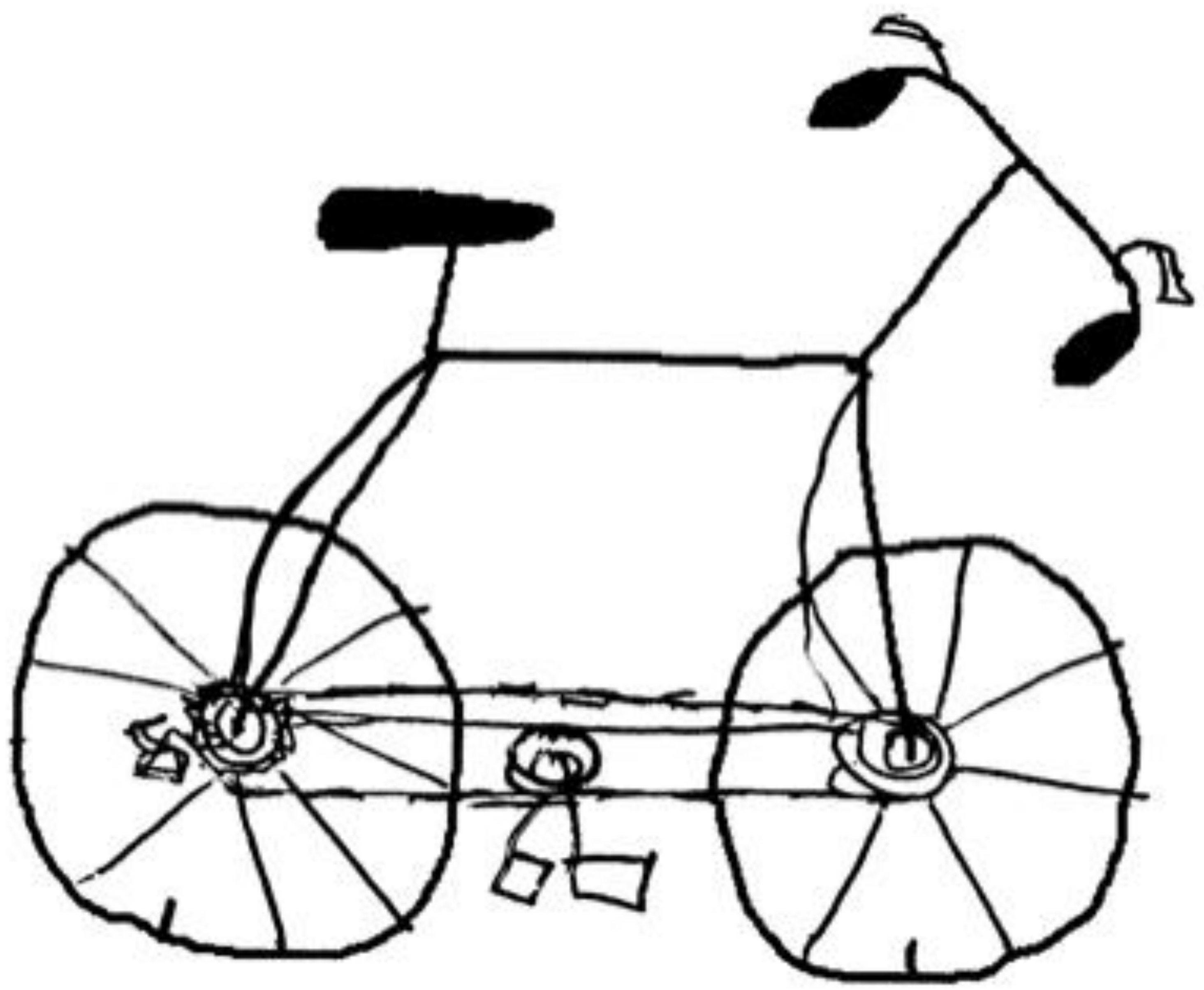
- The tendency for decision makers to start their decision process from a specific point of reference and subsequently make insufficient adjustments from the anchored point.
- Possible techniques to reduce this bias:
  - Document the rationale for a decision
  - Learn to anticipate and correct the bias
  - Develop your own expectations through experience

Exercise 2



https://www.liverpool.ac.uk/~rlawson/PDF_Files/L-M&C-2006.pdf

https://road.cc/content/blog/90885-science-cycology-can-you-draw-bicycle


Overconfidence bias

- The tendency for decision makers to overestimate their own abilities
- Two types here:
    - Overplacement
    - Illusion of explanatory depth
- From wikipedia: an incomplete list of events related or triggered by bias/overconfidence and a failing (safety) culture:
    - Chernobyl disaster
    - Sinking of the Titanic
    - Space Shuttle Challenger disaster
    - Space Shuttle Columbia disaster
    - Deepwater Horizon oil spill
    - Titan submersible implosion

A list of names for exercise 3

- Travis Kelce
- Olivia Rodrigo
- Blake Lively
- Rock Hudson
- Selena Gomez
- Kesha
- Taylor Swift
- Charles Barkley
- Cardi B
- Bruno Mars
- David Seaman
- Richard Grieco
- Sinbad
- Steve Winwood
- Megan Thee Stallion
- Barry White
- Steven Tyler
- Beyonce
- Millie Bobby Brown
- Lady Gaga
- Robert Duvall

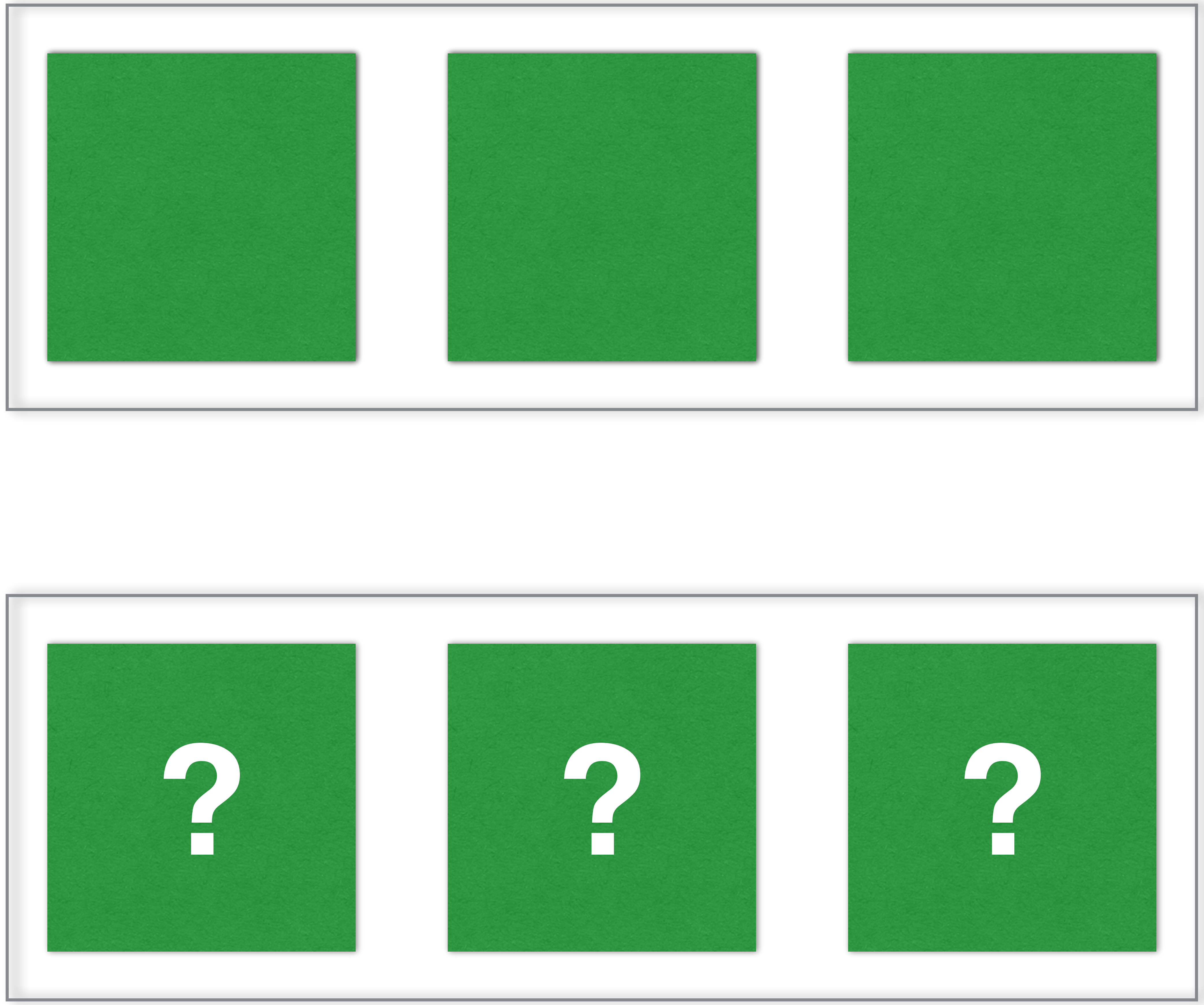
Exercise 3

Availability bias

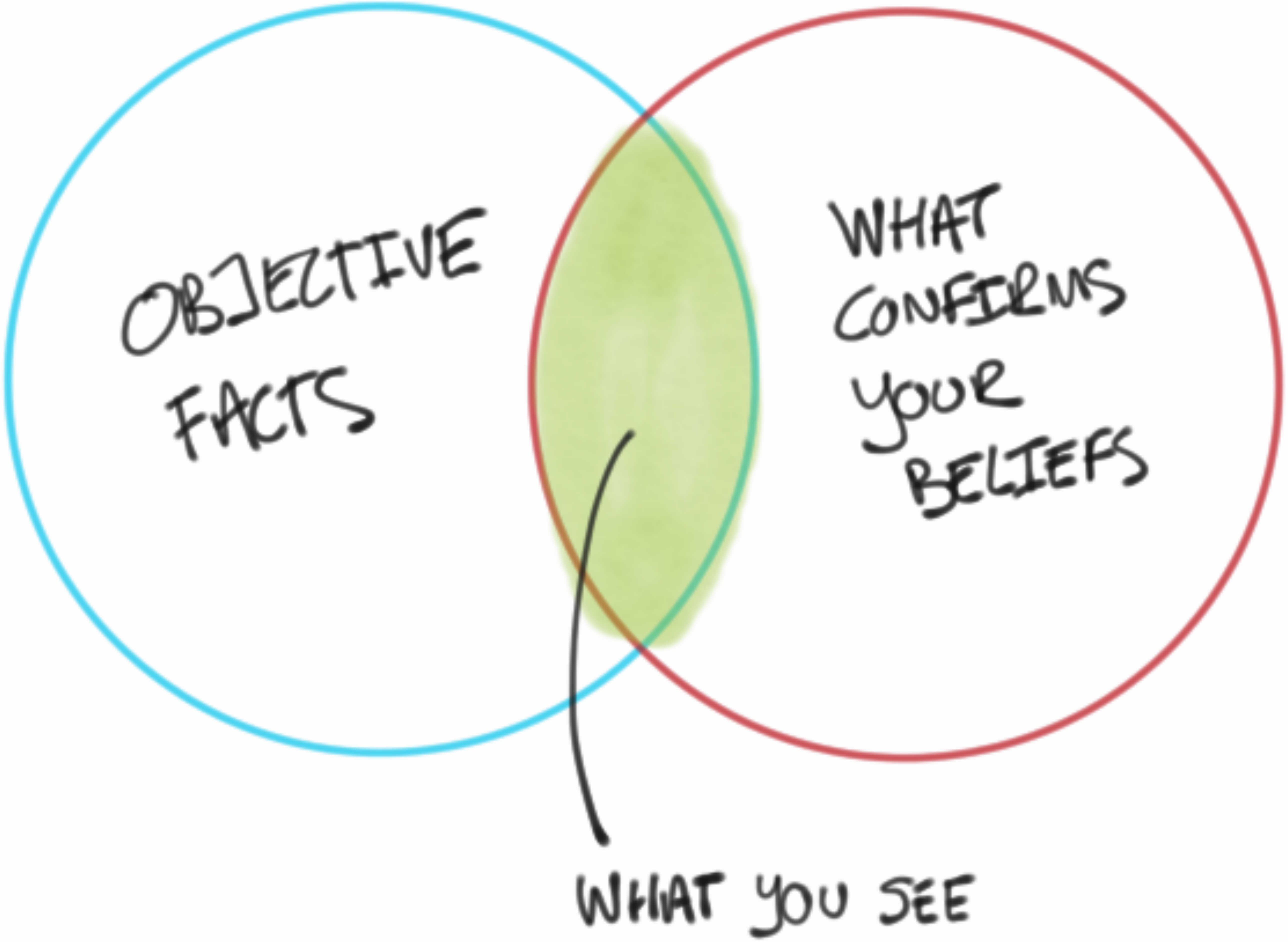
- The tendency for decision makers to consider information that is easily retrievable from memory as being more likely, more relevant, and more important for a judgment.
- Availability bias causes us to mistake the ease of recall for true frequency!!!
  - More generally: memory is weird and wonky!
- Disasters caused in part by availability bias
  - The "Jaws" Effect: Following the release of the movie Jaws in 1975, fear of shark attacks skyrocketed, leading to unnecessary, widespread shark hunts. People worry about shark attack, yet drowning is a far more likely danger at the beach.
  - Terrorism Overestimation: After a high-profile, televised terrorist attack, people tend to significantly overestimate the likelihood of another one occurring, due to the intense media coverage making the event readily available in their minds. e.g. Post 9-11 dip in air travel.
  - Rare disease fear, also some forms of vaccine hesitancy/avoidance

Exercise 4

Figure out the rule



Confirmation bias


OBJECTIVE FACTS
WHAT CONFIRMS YOUR BELIEFS
WHAT YOU SEE

Iran Air Flight 665 in 1988

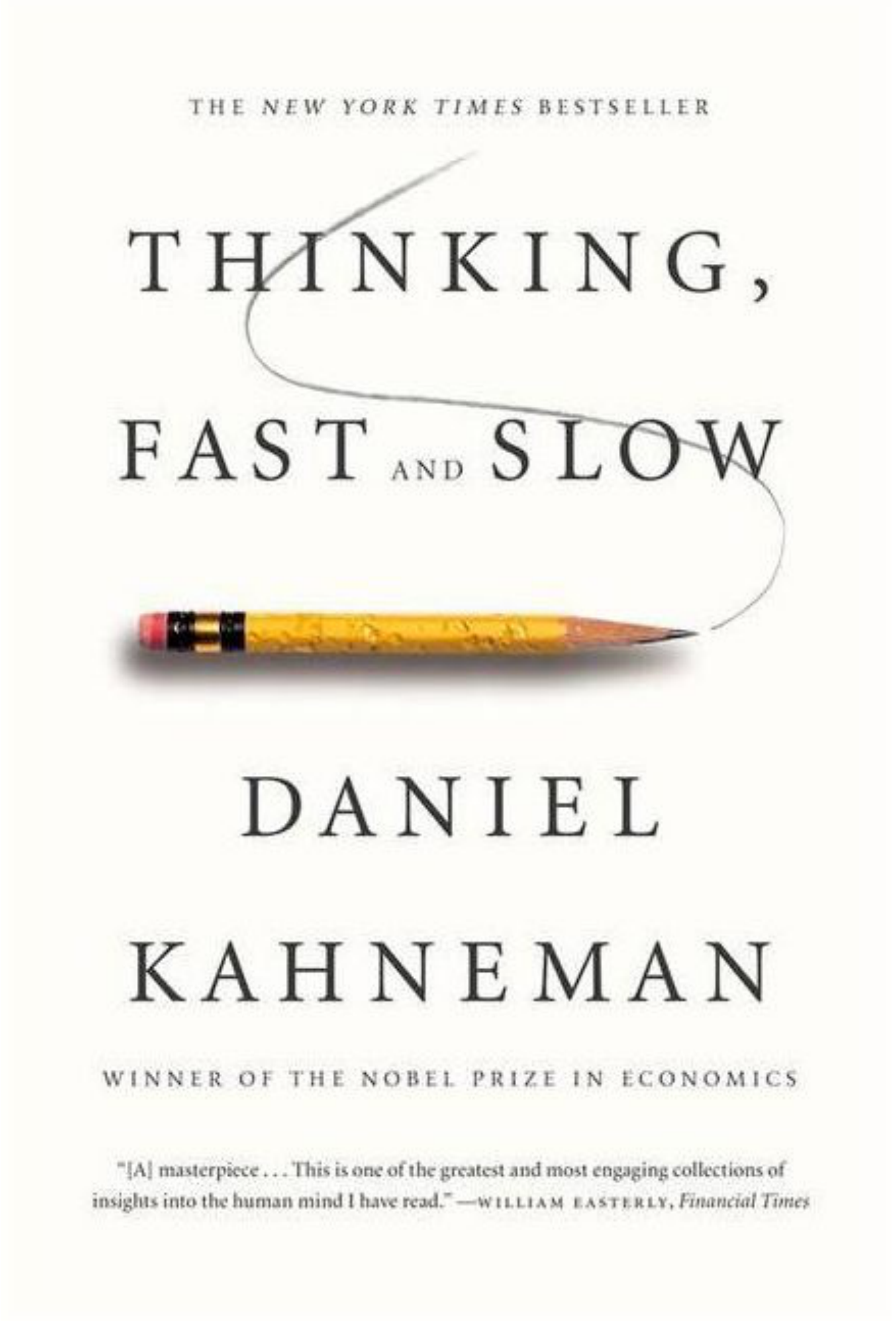
Confirmation bias contributes to a major disaster

- Scheduled short hop commercial flight across the gulf
- Iran - Iraq war in active hostilities
- 1 year before Iranian airplane sunk USS Stark
- USS Vincennes had already engaged gunboats that day
- Radar saw airliner at takeoff. Thought it was an attack because a technical glitch briefly confused the ID of aircraft with that of an Iranian F-14
- Had 7 min from radar detection to launch of SAM, lots of time to figure it out, lots of evidence against it being an F-14
- 290 people killed

Thinking Fast and Slow

One origin of biases

- System 1
  - Fast, effortless, feels involuntary
- System 2
  - Slower, effort+attention, feels like agency
- Not real brain systems or regions!
- Evolutionary needs for both, biases and illusions are often adaptive
- Kahneman + Tversky



How can biases affect analysis?

- Anchoring bias and Confirmation bias
  - False negatives: Reject a truth because its too far from expected
  - False positives: Accept a wrong result because it matched expected
- Illusion of explanatory depth
  - Confidently wrong, like ChatGPT is sometimes :)
- Availability bias
  - Decisions based on familiar results, novel (correct) ones matter less

Fighting biases

- Instruction, feedback, and practice
  - Learn how to test your hypotheses
  - Develop the habit of questioning yourself and your ideas aggressively
- Justification of judgements, accountability
  - Procedures that force you into questioning the basis of the decision
- Involving others if they are independent thinkers (beware groupthink!)

Jeff, Amy, and Tom had discussed Shoemaker Racing's situation the previous evening. This first season was a success from a racing standpoint, with the team's car finishing in the top five in twelve of the fifteen races it completed. As a result, the sponsorship offers critical to the team's business success were starting to come in. A big break had come two weeks ago after the Auto Club race, where the team scored its fourth first-place finish. Michelin tires had finally decided Shoemaker Racing deserved its sponsorship at the Sonoma race—worth a much needed $200,000—and was considering a full season contract for next year if the team's car finished in the top five in this race. The Michelin sponsorship was for $5 million a year, plus incentives. Jeff and Amy had gotten a favorable response from Michelin's racing program director last week when they presented their plans for next season, but it was clear that his support depended on the visibility they generated in this race.

"Jeff, we only have another hour to decide," Amy said over the phone. "At the end of the Auto Club race, and before we got the $200,000 from Michelin, we were $287,500 in the hole. If we withdraw now, we can get back half the $75,000 entry fee for the race. We will lose Michelin, they'll want $125,000 of their money back, and we'll end the season $250,000 in the hole. If we run and finish in the top five, we have Michelin in our pocket and can add another car next season. You know as well as I do, however, that if we run and lose another engine, we are back at square one next season. We will lose the Michelin sponsorship, and a blown engine is going to lose us our current oil contract for sure. No oil company wants a national TV audience to see a smoking car being dragged off the track with their name plastered all over it. The oil sponsorship is $2.5 million that we cannot live without. Think about it. Call Antonio and Tom if you want. But I need a decision within an hour."

Jeff hung up the phone and looked out of the window at the crisp fall sky. The temperature sign across the street flashed "39°F, 9:23 AM."

Jeff picked up the phone again and dialed. "Get Antonio Garcia for me." Jeff was calling to get his engine mechanic's opinion on whether they should run today or not. The data Tom put together indicated that temperature was not the problem, but Jeff wanted to get Antonio's direct assessment.

Broader questions

How can we be more right and less wrong?

Being aware of our biases; external review and checks to minimize

Thinking about tools; know their strengths and limitations

Worry about process and rigorous methods more than outcomes

Social factors influence decision making; good comm and procedures are essential

Be a systems level thinker! Stay curious! Question yourself and your team as much as possible